

Exercise 1

In the lecture we convinced ourselves that all maps are well-defined. As an inclusion, the first map is clearly injective.

Exactness at $\mathcal{O}_k^\times$ is clear, since $\rho(u) = 1 \in (\mathcal{O}_k/\mathfrak{m})^\times$ by definition means exactly $u \equiv 1 \pmod{\times \mathfrak{m}}$.

Exactness at $(\mathcal{O}_k/\mathfrak{m})^\times$: For $u \in \mathcal{O}_k^\times$ we have $(u) = (1)$, so $(u)\mathcal{P}_k(\mathfrak{m}) = \mathcal{P}_k(\mathfrak{m})$. Conversely, let $\alpha \in k^\times$, $(\alpha, \mathfrak{m}_0) = 1$ and suppose $(\alpha)\mathcal{P}_k(\mathfrak{m}) = \mathcal{P}_k(\mathfrak{m})$. That is, there is some $\varepsilon \in \mathcal{O}_k^\times$ s.t. $\varepsilon\alpha \equiv 1 \pmod{\times \mathfrak{m}}$, and $\rho(\alpha) = \rho(\varepsilon^{-1}) \in \rho(\mathcal{O}_k^\times)$.

Exactness at $\text{cl}_k(\mathfrak{m})$ is clear, since both kernel and image consist exactly of the classes of principal ideals coprime to $\mathfrak{m}$.

Finally, we show more generally, that if $\mathfrak{m} \mid \mathfrak{n}$, then the induced map $\text{cl}_k(\mathfrak{n}) \rightarrow \text{cl}_k(\mathfrak{m})$ is surjective: Let $\mathfrak{a}\mathcal{P}_k(\mathfrak{m}) \in \text{cl}_k(\mathfrak{m})$, for some ideal $\mathfrak{a} \in I_k(\mathfrak{m})$. By the approximation theorem, we find $\alpha \in k$ with $\alpha \equiv 1 \pmod{\times \mathfrak{m}}$ and $\nu_{\mathfrak{p}}(\alpha) = -\nu_{\mathfrak{p}}(\mathfrak{a})$ for all prime ideals $\mathfrak{p} \mid (\mathfrak{n}, \mathfrak{a})$. Then $(\alpha\mathfrak{a}, \mathfrak{n}) = 1$, and

$$\alpha\mathfrak{a}\mathcal{P}_k(\mathfrak{n}) \mapsto \alpha\mathfrak{a}\mathcal{P}_k(\mathfrak{m}) = \mathfrak{a}\mathcal{P}_k(\mathfrak{m}).$$

Exercise 2

If $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ is a short exact sequence of groups, then B is partitioned into $|C|$ cosets each of size $|A|$, hence $|B| = |A| \cdot |C|$. If $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow D \rightarrow 0$ is exact, let $K = \ker(C \rightarrow D) = \text{im}(A \rightarrow B)$. Then $0 \rightarrow A \rightarrow B \rightarrow K \rightarrow 0$ and $0 \rightarrow K \rightarrow C \rightarrow D \rightarrow 0$ are exact, and therefore

$$|A| \cdot |C| = |A| \cdot |K| \cdot |D| = |B| \cdot |D|.$$

(More generally, given any finite exact sequence, continuing in this way yields an equation of "alternating" cardinalities.)

Applying this to the exact sequence $0 \rightarrow \mathcal{O}_k^\times / (\mathcal{O}_k^\times \cap k_{\mathfrak{m}}^\times) \rightarrow \dots \rightarrow 0$ yields

$$[\mathcal{O}_k^\times : \mathcal{O}_k^\times \cap k_{\mathfrak{m}}^\times] \cdot |\text{cl}_k(\mathfrak{m})| = |\text{cl}_k| \cdot |(\mathcal{O}_k/\mathfrak{m})^\times|.$$

In particular, since the right hand side is finite, and groups cannot be empty, both cardinalities on the left hand-side are finite as well, and dividing by the first term yields the result.

Exercise 3

Let $k = \mathbb{Q}(\sqrt{d})$ and let $\tau_1 = \text{id}, \tau_2 : \overline{} \rightarrow \overline{}$ be the two embeddings associated to $\infty_{1/2}$. By Dirichlet, $\mathcal{O}_K^\times = \{\pm 1\} \times \varepsilon^{\mathbb{Z}}$, where wlog $\varepsilon > 1$. Clearly $-1 \not\equiv 1 \pmod{\times \mathfrak{m}}$. From $N_{k/\mathbb{Q}}(\varepsilon) = \varepsilon\bar{\varepsilon}$ we see $\text{sgn}(\bar{\varepsilon}) = N_{k/\mathbb{Q}}(\varepsilon)$. Hence if $N_{k/\mathbb{Q}}(\varepsilon) = 1$, then $\mathcal{O}_k^\times \cap k_{\mathfrak{m}}^\times = \varepsilon^{\mathbb{Z}}$ and the index in $\mathcal{O}_k^\times$ is 2. Otherwise, $\mathcal{O}_k^\times \cap k_{\mathfrak{m}}^\times = \varepsilon^{2\mathbb{Z}}$ and the index in $\mathcal{O}_k^\times$ is 4.

Further, since $\mathfrak{m}_0 = (1)$ we have $|(\mathcal{O}_k/\mathfrak{m})^\times| = |\{\pm 1\}^2| = 4$. The result now follows from exercise 2.

Exercise 1

We have

$$\frac{(\mathcal{O}_K/\mathfrak{f})^\times}{U} \cong \ker(\text{cl}_k(\mathfrak{f}) \rightarrow \text{cl}_k) \cong \ker(\text{Gal}(k(\mathfrak{f})/k) \rightarrow \text{Gal}(k(1)/k)) \cong \text{Gal}(k(\mathfrak{f})/k(1)),$$

where the first isomorphism follows from last week's sheet, the central one is induced by the Artin map (which is functorial), and the last is Galois theory.

Exercise 2

Let $\mathfrak{m} := q\mathbb{Z}\infty$. Then we know $\mathbb{Q}(\mathfrak{m}) = \mathbb{Q}(\zeta_q)$ and hence the Artin map induces an isomorphism

$$\text{cl}_{\mathbb{Q}}(\mathfrak{m}) \cong \text{Gal}(\mathbb{Q}(\zeta_q)/\mathbb{Q}) \cong (\mathbb{Z}/q\mathbb{Z})^\times \cong \mathbb{Z}/(q-1)\mathbb{Z}.$$

Let $H \subseteq \text{cl}_{\mathbb{Q}}(\mathfrak{m})$ be the unique subgroup of index 2. Then the existence theorem yields a unique abelian subfield $\mathbb{Q}(\zeta_q)/K/\mathbb{Q}$ with $\text{Gal}(K/\mathbb{Q}) \cong \text{cl}_{\mathbb{Q}}(\mathfrak{m})/H$, and ν ramifies in K only if $\nu \mid \mathfrak{m}$. Now the first condition implies that $K/\mathbb{Q}$ is quadratic, say $K = \mathbb{Q}(\sqrt{d})$ for some squarefree $d \in \mathbb{Z}$. Let p be a prime divisor of d . Then p ramifies in K , hence $p \mid q$. Hence either $d = -1$, or $d = \pm q$. Now 2 isn't allowed to ramify, so $d = -1$ is out, and we need the sign so that $\pm q \equiv 1 \pmod{4}$. Hence if $q \equiv 1 \pmod{4}$, we have $\mathbb{Q}(\sqrt{q}) \subseteq \mathbb{Q}(\zeta_q)$, and if $q \equiv 3 \pmod{4}$, then $\mathbb{Q}(\sqrt{-q}) \subseteq \mathbb{Q}(\zeta_q)$, hence

$$\mathbb{Q}(\sqrt{q}) \subseteq \mathbb{Q}(\sqrt{-q}, \sqrt{-1}) = \mathbb{Q}(\sqrt{-q})\mathbb{Q}(\zeta_4) \subseteq \mathbb{Q}(\zeta_q)\mathbb{Q}(\zeta_4) = \mathbb{Q}(\zeta_{4q}).$$

In both cases this implies the claim, since $\mathbb{Q}(\sqrt{q})$ is totally real.

Exercise 3

Let $m := [k : \mathbb{Q}_p]$. Then we showed in ANT that $k^\times \cong \mathbb{Z} \oplus \mathbb{Z}/(q-1)\mathbb{Z} \oplus \mathbb{Z}/p^a\mathbb{Z} \oplus \mathbb{Z}_p^m$ for a p -power q and non-negative integer a . Hence

$$k^\times / (k^\times)^n \cong \mathbb{Z}/n\mathbb{Z} \oplus \mathbb{Z}/(n, q-1)\mathbb{Z} \oplus \mathbb{Z}/(n, p^a)\mathbb{Z} \oplus (\mathbb{Z}_p/n\mathbb{Z}_p)^m.$$

The first three summands are clearly finite, hence it remains to consider $\mathbb{Z}_p/n\mathbb{Z}_p$. Write $n = p^v b$ with $v, b \geq 0$, $p \nmid b$. Then b is invertible in $\mathbb{Z}_p$, hence $\mathbb{Z}_p/n\mathbb{Z}_p \cong \mathbb{Z}_p/p^v\mathbb{Z}_p$, which is isomorphic to $\mathbb{Z}/p^v\mathbb{Z}$ by ANT. Putting everything together, we see that $(k^\times)^n$ has finite index in $k^\times$.

To see that $(k^\times)^n$ is open, it suffices to find an $N \in \mathbb{N}$ with $1 + \mathfrak{p}_k^N \subseteq (k^\times)^n$, then $(k^\times)^n = \bigcup_{\alpha \in (k^\times)^n} \alpha(1 + \mathfrak{p}_k^N)$ is a union of open sets, hence open. To do this, let $s > \frac{e}{p-1}$ and $N = s + v_k(n)$. Then

$$1 + \mathfrak{p}_k^N = \exp(n\mathfrak{p}_k^s) = \exp(\mathfrak{p}_k^s)^n = (1 + \mathfrak{p}_k^s)^n \subseteq (k^\times)^n.$$

Exercise 4

Let K/k be finite abelian of exponent p . By local class field theory, for $a \in k^\times$ we have $(a^p, K/k) = (a, K/k)^p = \text{id} \in \text{Gal}(K/k)$. Hence $(k^\times)^p \subseteq \ker((_, K/k))$. Conversely, if $(k^\times)^p \subseteq \ker((_, K/k))$, then since the Artin map is surjective, $\text{Gal}(K/k)$ has exponent (dividing) p . Hence the finite abelian extensions of exponent p correspond exactly to the subgroups $(k^\times)^p \subseteq H \subsetneq k^\times$ (they are automatically open).

By the same calculation as in exercise 3, for $p \neq 2$ we have $k^\times / (k^\times)^p \cong \mathbb{Z}/p\mathbb{Z} \oplus \mathbb{Z}/p\mathbb{Z}$. This has $\frac{p^2-1}{p-1} = p+1$ subgroups of order p , and the trivial subgroup, so there are $p+2$ finite abelian extensions of $\mathbb{Q}_p$ of exponent p . For $p = 2$, the same method yields $k^\times / (k^\times)^2 \cong (\mathbb{Z}/2\mathbb{Z})^3$, which has 7 subgroups of order 2, and $\frac{1}{3}\binom{7}{2} = 7$ subgroups of order 4, so there are $1 + 7 + 7 = 15$ finite abelian extensions of $\mathbb{Q}_2$ of exponent 2.

Exercise 1

Since $k^\times \subseteq J_k$ are Hausdorff topological groups, it suffices to show that $\{1\} \subseteq k^\times$ is open, i.e. that there exists an open neighbourhood $1 \in U \subseteq J_k$ with $U \cap k^\times = \{1\}$. Let S_∞ be the set of real places of k , and set $U_\varepsilon := \prod_{v \in S_\infty} B_\varepsilon(1) \times \prod_{v \notin S_\infty} \mathcal{U}_v$ for $\varepsilon > 0$. By definition of the product topology, U_ε is open. Now let $\alpha \in k^\times$. Then $\alpha \in U_\varepsilon$ implies $\alpha \in \mathcal{O}_{k_v}$ for the finite places, i.e. $\alpha \in \mathcal{O}_k$, and $|\alpha - 1|_v < \varepsilon$ for every infinite place $v \in S_\infty$. But $\mathcal{O}_k$ embeds as a lattice in $\mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$, which is in particular discrete. So for ε small enough, no $\alpha \in \mathcal{O}_k$ satisfies these conditions.

Exercise 2

Let cl_k be generated by the classes of $\mathfrak{p}_1, \dots, \mathfrak{p}_r$. Then set $S := S_\infty \cup \{\mathfrak{p}_1, \dots, \mathfrak{p}_r\}$. Let $\alpha \in J_k$ and denote by $\varphi : J_k/k^\times \mathcal{U}_k \rightarrow \text{cl}_k$ the isomorphism from exercise 3. Then by construction, write $\varphi(\alpha) = x \prod \mathfrak{p}_i^{e_i}$ for some $e_i \in \mathbb{Z}$ and $x \in K^\times$. Set $\beta = \alpha x^{-1}$, we have to show $\beta \in J_k^S$. So let $v \notin S$, then v is finite. We have $v(\alpha) = v(\varphi(\alpha)) = v(x)$ since $v \notin \{\mathfrak{p}_1, \dots, \mathfrak{p}_r\}$, but $\varphi(\alpha)$ and x only differ by factors of the $\mathfrak{p}_i$. Hence $v(\beta) = v(\alpha) - v(x) = 0$ and $\beta \in \mathcal{U}_v$, as required.

From $J_k = J_k^S k^\times$ it immediately follows that $C_k^S = J_k^S k^\times / k^\times = J_k / k^\times = C_k$.

Exercise 3

(a) Consider the content map $\bar{c} : J_k \xrightarrow{c} I_k \twoheadrightarrow \text{cl}_k$. The kernel of $\bar{c}$ is $c^{-1}(k^\times) = k^\times \ker c = k^\times \mathcal{U}_k$.

(b) Consider the commutative diagram defining the Artin map on J_k (after theorem 1.45). Let $\mathfrak{m} = 1$ and $K = k(1)$. We know that ideal-theoretically, K corresponds to $P_k(\mathfrak{m})$. Also, the top arrow is induced by the content, so it is exactly the map in (a). Finally, since $\mathfrak{m} = 1$ the left vertical arrow is the identity. Under all these simplifications, the Artin map reduces to the composition

$$J_k \twoheadrightarrow J_k/k^\times \mathcal{U}_k \xrightarrow[\cong]{\bar{c}} \text{cl}_k \xrightarrow[\cong]{(_, K/k)} \text{Gal}(K/k).$$

Since the last two arrows are isomorphisms, the kernel is just the kernel of the first one, which is obviously $k^\times \mathcal{U}_k$.

Exercise 4

(a) It's clear that $\mathfrak{p}$ ramifies in L/k if and only if $\sigma(\mathfrak{p})$ ramifies in $\widehat{\sigma}(L)/k$. So $\widehat{\sigma}(L)/k$ is unramified outside $\sigma(\mathfrak{m})$, and the Artin map is well-defined on $I_k(\sigma(\mathfrak{m}))$. Now let $\mathfrak{p} \in I_k(\mathfrak{m})$ be a prime ideal, and $\tau := (\mathfrak{p}, L/k) \in \text{Gal}(L/k)$ its Frobenius. Then τ is uniquely determined by $\tau(\alpha) \equiv \alpha^f \pmod{\mathfrak{p}}$, where $f := N(\mathfrak{p})$. Let $\alpha \in \widehat{\sigma}(L)$, then $\tau(\widehat{\sigma}^{-1}\alpha) \equiv (\widehat{\sigma}^{-1}\alpha)^f \equiv \widehat{\sigma}^{-1}(\alpha^f) \pmod{\mathfrak{p}}$, hence $\widehat{\sigma}\tau\widehat{\sigma}^{-1}(\alpha) \equiv \widehat{\sigma}\widehat{\sigma}^{-1}(\alpha^f) \equiv \alpha^f \pmod{\sigma(\mathfrak{p})}$, which precisely means $(\sigma(\mathfrak{p}), \widehat{\sigma}(L)/k) = \widehat{\sigma}(\mathfrak{p}, L/K)\widehat{\sigma}^{-1}$. Since the prime ideals generate the fractional ideals, this identity automatically holds for all fractional ideals.

In particular, using that L is defined mod $\mathfrak{m}$, i.e. $P_k(\mathfrak{m}) \subseteq \ker(_, L/k)$, we see for $\sigma(\mathfrak{a}) \in \sigma(P_k(\mathfrak{m})) = P_k(\sigma(\mathfrak{m}))$ that $(\sigma(\mathfrak{a}), \widehat{\sigma}(L)/k) = \widehat{\sigma}(\mathfrak{a}, L/k)\widehat{\sigma}^{-1} = \widehat{\sigma}\widehat{\sigma}^{-1} = 1$, i.e. $P_k(\sigma(\mathfrak{m})) \subseteq \ker(_, \widehat{\sigma}(L)/k)$, as desired.

(b) Since $\ker(_, \widehat{\sigma}(k_H)/k) = \sigma(\ker(_, k_H/k))$, we see that $\widehat{\sigma}(k_H)$ corresponds to $\sigma(H)$.

Now k_H/F is galois if and only if $\widehat{\sigma}(k_H) = k_H$ for all $\widehat{\sigma}$, if and only if $\sigma(H) = H$, for all $\sigma \in G$, if and only if $\sigma(H) \subseteq H$ for all $\sigma \in G$.

Exercise 1

Let $\mathfrak{m} := \mathfrak{q}$ be a prime of k . By exercise 1.28, $[k(\mathfrak{m}) : H] = \frac{\omega(1)}{\omega(\mathfrak{m})} \varphi_k(\mathfrak{m}) = \frac{\omega(1)}{\omega(\mathfrak{m})} (N(\mathfrak{q}) - 1)$. If $\mathfrak{q}$ is split in $k/\mathbb{Q}$, then $N(\mathfrak{q}) =: q$ is a prime number, and by class field theory there exists a field $H \subseteq K \subseteq k(\mathfrak{m})$ with the desired properties (the conductor must divide $\mathfrak{q}$ and cannot be trivial) if and only if $p \mid [k(\mathfrak{m}) : H]$. Hence it suffices to show that there are infinitely many primes $q \equiv 1 \pmod{p}$ that split in k .

If $p = 2$ this is clear, so now let p be odd and $k = \mathbb{Q}(\sqrt{d})$, $d < 0$. q splits in k if and only if $1 = \left(\frac{d}{q}\right)$. Write $d = p^n e$ with $p \nmid e$, then by quadratic reciprocity, q splits in k if and only if $\left(\frac{e}{q}\right) = (-1)^{n(p-1)(q-1)/4}$, which is a congruence condition $\pmod{4e}$. By the Chinese Remainder Theorem and Dirichlet's Theorem on Arithmetic Progressions, there are infinitely many primes $q \equiv 1 \pmod{p}$ satisfying this condition, hence infinitely many field extensions as required.

Exercise 2

Clearly it suffices to prove (b). Recall that I_G is the free abelian group generated by $\sigma - 1$, $1 \neq \sigma \in G$. Hence set $\varphi : I_G \rightarrow G/G'$, $\sigma - 1 \mapsto \sigma$ and extend linearly. Now

$$\varphi((\sigma - 1)(\tau - 1)) = \varphi((\sigma\tau - 1) - (\sigma - 1) - (\tau - 1)) = \sigma\tau\sigma^{-1}\tau^{-1}G' = 1,$$

so $I_G^2 \subseteq \ker \varphi$ and we get a well-defined morphism $\bar{\varphi} : I_G/I_G^2 \rightarrow G/G'$.

Conversely, let $\psi : G \rightarrow I_G/I_G^2$, $\sigma \mapsto \sigma - 1 + I_G^2$. For $\sigma, \tau \in G$, we have

$$\psi(\sigma\tau) = \sigma\tau - 1 + I_G^2 = (\sigma - 1) + (\tau - 1) + (\sigma - 1)(\tau - 1) = \psi(\sigma) + \psi(\tau).$$

Thus ψ is a group homomorphism. Since I_G/I_G^2 is abelian, it factors through $\bar{\psi} : G/G' \rightarrow I_G/I_G^2$. Clearly $\bar{\varphi}$ and $\bar{\psi}$ are mutually inverse, hence $I_G/I_G^2 \cong G/G' = G^{\text{ab}}$.

Exercise 3

Let $G = \{\pm 1\}$, $A = \mathbb{Z}/2\mathbb{Z}$ with the trivial G -action and $B = \mathbb{Z}/4\mathbb{Z}$ with the action $-1 \cdot b = -b = 3b$. Then $B^G = \{0, 2\} \cong \mathbb{Z}/2\mathbb{Z}$, and $A \otimes B^G \cong (\mathbb{Z}/2\mathbb{Z}) \otimes (\mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ (recall that, in general, multiplication induces an isomorphism $C_n \otimes C_m \cong C_{\gcd(n,m)}$) with the non-trivial element $1 \otimes 2$. Its image in $A \otimes B = (\mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z}/4\mathbb{Z})$ is $1 \otimes 2 = 2(1 \otimes 1) = 2 \otimes 1 = 0$, so the canonical map $A^G \otimes B^G \rightarrow (A \otimes B)^G$ is the zero map. Since $1 \otimes -1 = 3(1 \otimes 1) = 1 \otimes 1$, we have $(A \otimes B)^G = A \otimes B \cong \mathbb{Z}/2\mathbb{Z}$, so the map is not surjective either, even if the action of G on A is trivial.

Exercise 4

(1) $\Rightarrow$ (4): Let $0 \rightarrow A \rightarrow B \rightarrow X \rightarrow 0$ be exact, with X projective. Thus there exists a morphism making the following diagram commute:

$$\begin{array}{ccccc} & & X & & \\ & \swarrow \exists s & \downarrow \text{id} & & \\ B & \longrightarrow & X & \longrightarrow & 0 \end{array}$$

which is exactly the required splitting.

(4) $\Rightarrow$ (3): The morphism $\pi : B := \bigoplus_{x \in X} \mathbb{Z}[G] \rightarrow X$, $x \mapsto x$ is clearly surjective. By assumption let $s : X \rightarrow A$ be a splitting of the short exact sequence $0 \rightarrow \ker \pi \rightarrow B \rightarrow X \rightarrow 0$, then define $\varphi : \ker \pi \oplus X \rightarrow B$, $(a, x) \mapsto a + s(x)$. Let $(a, x) \in \ker \varphi$, then $0 = \pi(\varphi(a, x)) = \pi(s(x)) = x$, hence

also $a = \varphi(a, 0) = 0$. Conversely, let $b \in B$. Then $\pi(b - \sigma(\pi(b))) = 0$, so $b - \sigma(\pi(b)) = a$ for some $a \in \ker \pi$. But that exactly means $b = \varphi(a, \sigma(\pi(b)))$, so φ is an isomorphism and X is direct summand of a free G -module.

(3) $\Rightarrow$ (2): Let $X \oplus Y = Z$ with Z a free G -Module. The result follows immediately from $\text{Hom}(Z, _)$ being exact and $\text{Hom}(Z, _) = \text{Hom}(X, _) \times \text{Hom}(Y, _)$.

(2) $\Rightarrow$ (1): Let $g : B \rightarrow C$ be surjective, and $h : X \rightarrow C$ any morphism. Then by assumption $\text{Hom}(X, B) \rightarrow \text{Hom}(X, C) \rightarrow 0$ is surjective, so there is a preimage $s : X \rightarrow B$ of h .

Exercise 1

(a) Let $a \in A^G$. Then $gi(a) = i(ga) = i(a) \in B^G$, so the induced sequence is well-defined. It is clear that $i|_{A^G}$ is injective, since i is, and that $j|_{B^G} \circ i|_{A^G} = 0$. Conversely, let $b \in \ker j|_{B^G}$. Since the original sequence is exact, there is $a \in A$ with $i(a) = b$, we have to show $a \in A^G$. For $g \in G$, $i(ga) = gi(a) = gb = b = i(a)$ and the claim follows from the injectivity of i .

(b) By (a) and the long exact cohomology sequence, exactness everywhere but at C^G and $H^1(G, C)$ is clear, and the latter follows directly from $\text{im } \delta_G = \text{im } \delta_0$. For $b \in B^G$ we have $\delta_G(j(b)) = \delta_0(j_*(\bar{b})) = 0$. Conversely, let $c \in \ker \delta_G$, i.e. $\delta_0(\bar{c}) = 0$. By exactness of the long exact cohomology sequence, there is $b \in B^G$ such that $j_*(\bar{b}) = \bar{c}$. Hence $c = j(b) + N_G c'$ for some $c' \in C$. Let $b' \in B$ be a preimage of c' , then $c = j(b) + N_G j(b') = j(b + N_G b') \in \text{im } j|_{B^G}$.

Exercise 2

Let $L = \mathbb{Q}(\sqrt{d})$ with $0, 1 \neq d \in \mathbb{Z}$ squarefree. Let $h = 1$ if $d \equiv 1 \pmod{4}$ and $h = 2$ otherwise. Denote $G = \{1, \sigma\}$.

$H^q(G, \mathcal{O}_L)$: Let $\alpha = x + y\sqrt{d} \in \mathcal{O}_L$. Then $N_G \alpha = 2x$, so $N_G \mathcal{O}_L = \mathbb{Z}\sqrt{d}$ and $N_G \mathcal{O}_L = h\mathbb{Z}$. Furthermore

$$I_G \mathcal{O}_L = \{\sigma\alpha - \alpha \mid \alpha \in \mathcal{O}_L\} = \{2y\sqrt{d} \mid y \in \frac{h}{2}\mathbb{Z}\} = h\mathbb{Z}\sqrt{d}$$

and $\mathcal{O}_L^G = \mathcal{O}_L \cap \mathbb{Q} = \mathbb{Z}$, hence

$$H^{-1}(G, \mathcal{O}_L) \cong N_G \mathcal{O}_L / I_G \mathcal{O}_L \cong \begin{cases} 0 & \text{if } d \equiv 1 \pmod{4} \\ \mathbb{Z}/2\mathbb{Z} & \text{otherwise.} \end{cases} \quad H^0(G, \mathcal{O}_L) \cong \mathcal{O}_L^G / N_G \mathcal{O}_L \cong \begin{cases} 0 & \text{if } d \equiv 1 \pmod{4} \\ \mathbb{Z}/2\mathbb{Z} & \text{otherwise.} \end{cases}$$

$H^q(G, \mu_L)$: On μ_L we have that N_G is trivial, $I_G \mathcal{O}_L^\times = \mu_L^2$ and $(\mu_L^\times)^G = \mu_L^\times \cap \mathbb{Q} = \{\pm 1\}$. Hence $H^{-1}(G, \mu_L^\times) \cong \mathbb{Z}/2\mathbb{Z} \cong H^0(G, \mu_L^\times)$.

$H^q(G, \mathcal{O}_L^\times)$: If $d < 0$, then $\mathcal{O}_L^\times = \mu_L$. So let $d > 0$, $\varepsilon \in \mathcal{O}_L^\times$ a fundamental unit, and assume $N(\varepsilon) = 1$, so that $N_G = 2$. We have $I_G \mathcal{O}_L = \langle \varepsilon^{\sigma-1} \rangle = \langle \varepsilon^{-2} \rangle$ and $(\mathcal{O}_L^\times)^G = \{\pm 1\}$ as before, leading to $H^{-1}(G, \mathcal{O}_L^\times) \cong (\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}) / (0 \times 2\mathbb{Z}) \cong (\mathbb{Z}/2\mathbb{Z})^2$ and $H^0(G, \mathcal{O}_L^\times) \cong \mathbb{Z}/2\mathbb{Z}$.

Now suppose $N(\varepsilon) = -1$. Then $N_G \mathcal{O}_L^\times = \{\pm 1\} = (\mathcal{O}_L^\times)^G$, so $H^0(G, \mathcal{O}_L^\times) = 0$ and $N_G \mathcal{O}_L^\times = \langle \pm \varepsilon^2 \rangle$ while $I_G \mathcal{O}_L^\times = \langle -\varepsilon^{-2} \rangle$, hence $H^{-1}(G, \mathcal{O}_L^\times) \cong \mathbb{Z}/2\mathbb{Z}$.

$H^q(G, I_L)$: We have $I_L \cong \bigoplus_{\mathfrak{p}} \mathbb{Z} \cong \bigoplus_p \bigoplus_{\mathfrak{p}|p} \mathbb{Z}$, where p denotes primes in $\mathbb{Z}$ and $\mathfrak{p}$ primes in L . Since the last decomposition respects the G -module structure, we have $H^q(G, I_L) \cong \bigoplus_p H^q(G, \bigoplus_{\mathfrak{p}|p} \mathbb{Z})$. If p does not split in L , then the action of G is trivial on $I_p := \bigoplus_{\mathfrak{p}|p} \mathbb{Z} = \mathfrak{p}\mathbb{Z}$, so $H^{-1}(G, I_p) = 0$ and $H^0(G, I_p) = \mathbb{Z}/2\mathbb{Z}$. If p splits, then σ permutes the factors of $I_p = \mathbb{Z} \oplus \mathbb{Z}$, so that $N_G I_p = \langle (1, -1) \rangle$, $N_G I_p = \langle (1, 1) \rangle = I_p^G$ as well as

$$I_G I_p = \{(1 - \sigma)(x, y) \mid x, y \in \mathbb{Z}\} = \{(x - y, y - x) \mid x, y \in \mathbb{Z}\} = \langle (1, -1) \rangle,$$

hence $H^{-1}(G, I_L) = 0 = H^0(G, I_L)$. Putting things together, we find

$$H^{-1}(G, I_L) = 0, \quad H^0(G, I_L) \cong \bigoplus_{p \text{ non-split in } L} \mathbb{Z}/2\mathbb{Z}.$$

Exercise 3

By Hilbert's Theorem 90 (see ANT1, Sheet 14, Exercise 4) we have

$$H^{-1}(G, L^\times) \cong N_G L^\times / I_G L^\times = \{a \in L^\times \mid N_{L/K}(a) = 1\} / \{\frac{\sigma(b)}{b} \mid b \in L^\times\} = 0.$$

Exercise 4

(a) Since $(p, m) = 1$ we know that f is separable. Let $\alpha_1, \dots, \alpha_m$ be the pairwise different roots of f . Then $(\frac{\alpha_i}{\alpha_1})^m = \frac{a}{a} = 1$, so the $\frac{\alpha_i}{\alpha_1}$ are pairwise different m -th roots of unity. This immediately implies $\mu_m = \{\frac{\alpha_i}{\alpha_1} \mid i = 1, \dots, m\} \subseteq L$.

(b) It is clear that x is a well-defined map, we have to show that it is a crossed homomorphism. So let $\sigma, \tau \in G$. Then

$$x(\sigma\tau) = \frac{\sigma\tau(\omega)}{\omega} = \frac{\sigma\tau(\omega)}{\sigma(\omega)} \frac{\sigma(\omega)}{\omega} = \sigma(x(\tau))x(\sigma),$$

so $\bar{x}$ is an element of $H^1(G, \mu_m)$. Now let $\omega' \in L$ be another element with $\omega'^m = a$, then $\omega' = \zeta\omega$ for some $\zeta \in \mu_m$, and $\frac{\sigma(\omega')}{\omega'} = \frac{\sigma(\zeta\omega)}{\zeta\omega} = \frac{\sigma(\zeta)}{\zeta} \frac{\sigma(\omega)}{\omega}$ with $\sigma \mapsto \frac{\sigma(\zeta)}{\zeta}$ being a 1-coboundary. This shows that the class $\bar{x} \in H^1(G, \mu_m)$ is independent of the choice of ω .

Exercise 1

(a) $A^G \subseteq A^U$ is clear, so it remains to show $N_G A \subseteq N_U A$. Let $\sigma_1, \dots, \sigma_k$ be a system of representatives for $U \setminus G$, and set $N_{U \setminus G} = \sum_{i=1}^k \sigma_i$. Then $N_G = \sum_{g \in G} g = \sum_{i=1}^k \sum_{u \in U} u \sigma_i = N_U N_{U \setminus G}$, which immediately implies the claim.

(b) Let $\bar{c} \in H^0(G, C)$. Let $\bar{b} \in H^0(G, B)$ with $jb = c$, and $\bar{a} \in H^1(G, A)$ with $i_* a = \partial b$, so that $\delta(\bar{c}) = \bar{a}$. Now clearly $i(\text{res}_1(a)) = \text{res}_1(\partial b) = \partial_U b$, so $\delta(\text{Res}_0(\bar{c})) = \text{Res}_1(\bar{a}) = \text{Res}_1(\delta(\bar{c}))$

Exercise 2

Consider the diagram

$$\begin{array}{ccccc}
 U^{\text{ab}} & \xrightarrow{\cong} & I_U/I_U^2 = H^{-1}(U, I_U) & \xleftarrow{\delta} & H^{-2}(U, \mathbb{Z}) \\
 \downarrow \varphi & & \downarrow i_* & \swarrow \delta & \downarrow \text{Kor}_{-2} \\
 & & H^{-1}(U, I_G) & & \\
 & & \downarrow \text{Kor}_{-1} & & \\
 G^{\text{ab}} & \xrightarrow{\cong} & I_G/I_G^2 = H^{-1}(G, I_G) & \xleftarrow{\delta} & H^{-2}(G, \mathbb{Z})
 \end{array}$$

Here, the trapezoid on the right commutes by definition of corestriction with the short exact sequence $0 \rightarrow I_G \rightarrow \mathbb{Z}[G] \rightarrow \mathbb{Z} \rightarrow 0$, and the commutativity of the triangle follows from naturality of the long exact cohomology sequence. The big rectangle commutes by definition of φ and Exercise 4.2, hence the left square also commutes. Hence for $g \in U$, the image of $\varphi(gU')$ is given by

$$\text{Kor}_{-1}(i_*(g - 1 + I_U^2)) = \text{Kor}_{-1}(g - 1 + I_U I_G) = g - 1 + I_G^2,$$

which maps to gG' in G^{ab} . Hence $\varphi(gU') = gG'$.

Exercise 3

By definition, $\text{Kor}_0 : H^0(U, L^\times) = F^\times / N_{L/F} L^\times \rightarrow K^\times / N_{L/K} L^\times = H^0(G, L^\times)$ is given by $\alpha N_{L/F} L^\times \mapsto N_{G/U}(\alpha) N_{L/K} L^\times$, where $N_{G/U} \in \mathbb{Z}[G]$ is the sum of a system of representatives of G/U . Let $\sigma_1, \dots, \sigma_k \in \text{Gal}(L/K)$ be such a system, then the $\sigma_i|_F$ are $|G/U| = [F : K]$ pairwise different morphisms $F \rightarrow \bar{K}$, hence for $\alpha \in F^\times$ we have $N_{F/K}(\alpha) = \prod_{i=1}^k \sigma_i(\alpha) = N_{G/U}(\alpha)$ as required. The argument for the trace is exactly the same, reading everything additively.

Exercise 4

Given $x : G^q \rightarrow A$, let $f^*x : (G')^q \rightarrow f^*A$, $(\sigma_1, \dots, \sigma_q) \mapsto x(f(\sigma_1), \dots, f(\sigma_q))$, we want to show that this induces a morphism $f^* : H^q(G, A) \rightarrow H^q(G', f^*A)$, i.e. we have to show that the diagram

$$\begin{array}{ccc}
 A_q & \xrightarrow{\partial} & A_{q+1} \\
 \downarrow f^* & & \downarrow f^* \\
 (f^*A)_q & \xrightarrow{\partial} & (f^*A)_{q+1}
 \end{array}$$

commutes for $q \geq 0$. But this is clear from the explicit formulas for ∂ . Now composition with $g_* : H^q(G', f^*A) \rightarrow H^q(G', A')$ gives the desired morphism $(f, g)^*$.

If $f : H \hookrightarrow G$ is the inclusion of a subgroup, then for $x \in A_q$, $f^*x = x|_{H^q}$ by definition, which induces the restriction. Now let $H \subseteq G$ be a normal subgroup, $f : G \rightarrow G/H$ the projection and $g : A^H \rightarrow A$ the inclusion. It is clear that f and g are compatible. By definition, (f, g) is induced by the map of cochains $(x : (G/H)^q \rightarrow A^H) \mapsto (G^q \rightarrow A, \sigma \mapsto g(x(f(\sigma))))$, which is just like inflation is defined.

Exercise 1

(a) Let $\bar{c} \in H^q(G, C)$, $c \in C_q$. Let $b \in B_q$ with $j_*b = c$ and $a \in A_{q+1}$ with $i_*a = \partial b$, so that $\delta(\bar{c}) = \bar{a}$ and $\text{res}_{q+1}(\delta(\bar{c})) = \bar{a}|_{g^q}$. On the other hand, we have $\mathfrak{a}_*b|_{g^q} = (j_*b)|_{g^q} = c|_{g^q}$ and $i_*a|_{g^q} = (\partial b)|_{g^q} = \partial(b|_{g^q})$, which means $\delta(\text{res}_q \bar{c}) = \bar{a}|_{g^q}$.

(b) Follows directly from the uniqueness of restrictions in last week's exercises.

Exercise 2

$N_G a_1(\sigma) = \sum_{\tau \in G} \tau a_1(\sigma) = \sum_{\tau \in G} a_1(\sigma\tau) - a_1(\tau) = 0$ shows $a_1(\sigma) \in N_G A$, i.e. $a_1(\sigma) + I_G A \in H^{-1}(G, A)$. Now let a_1 be a 1-coboundary, that is $a_1(\sigma) = \sigma x - x$ for some $x \in A$. This says exactly that $a_1(\sigma) \in I_G A$, i.e. $a_1(\sigma) + I_G A$ is trivial. Hence $a_1(\sigma) + I_G A$ only depends on the class $\bar{a}_1 \in H^1(G, A)$.

Exercise 3

Let $\rho \in G$. Using the 2-cocycle condition, we have

$$\rho a_0 = \sum_{\tau \in G} \rho a_2(\tau, \sigma) = \sum_{\tau \in G} a_2(\rho\tau, \sigma) + a_2(\rho, \tau) - a_2(\rho, \tau\sigma) = \sum_{\tau \in G} a_2(\rho\tau, \sigma) = a_0,$$

hence $a_0 \in A^G$ and $\bar{a}_0 \in H^0(G, A)$ is well-defined. Now let a_2 be a 2-coboundary, that is $a_2(\sigma, \tau) = \sigma y(\tau) - y(\sigma\tau) + y(\sigma)$ for some $y : G \rightarrow A$; we have to show that a_0 is a norm. Plugging in this formula yields

$$a_0 = \sum_{\tau \in G} a_2(\tau, \sigma) = \sum_{\tau \in G} \tau y(\sigma) - y(\tau\sigma) + y(\tau) = \sum_{\tau \in G} y(\sigma) = N_G y(\sigma).$$

Exercise 4

(a) Let $u' : G \rightarrow E$ be another section of π , then $\pi(u'_x u_x^{-1}) = x x^{-1} = 1$, so by exactness $u'_x = u_x a'$ for some $a' \in A$. Then $u'_x a(u'_x)^{-1} = u_x a' a(a')^{-1} u_x^{-1} = u_x a u_x^{-1}$ since A is abelian.

(b) Follows directly from equation (2).

(c) As above, $\pi(u_x u_y u_{xy}^{-1}) = 1$, so there exists some $f(x, y) \in A$ with $u_x u_y = f(x, y) u_{xy}$.

(d) Let $x, y, z \in G$. One computes

$$\begin{aligned} f(x, y) f(xy, z) u_{xyz} &= f(x, y) u_{xy} u_z = (u_x u_y) u_z = u_x (u_y u_z) \\ &= u_x f(y, z) u_{yz} = x f(y, z) u_{xyz} = x f(y, z) f(x, yz) u_{xyz} \end{aligned}$$

and cancelling u_{xyz} gives exactly the 2-cocycle condition.

(e) Let $f' \in Z^2(G, A)$ be the cocycle obtained from u' . Then $u' \cdot u^{-1}$ has values in A , i.e. is a 1-cochain, and

$$\begin{aligned} f(x, y) \partial_2(u' u^{-1})(x, y) &= f(x, y) x (u'_y u_y^{-1}) u_{xy} (u'_{xy})^{-1} u'_x u_x^{-1} \\ &= u_x u'_y u_y^{-1} u_x^{-1} f(x, y) u_{xy} (u'_{xy})^{-1} u'_x u_x^{-1} \\ &= u_x u'_y (u'_{xy})^{-1} u'_x u_x^{-1} = u_x (u'_x)^{-1} f'(x, y) u'_x u_x^{-1} \\ &= x x^{-1} f'(x, y) = f'(x, y). \end{aligned}$$